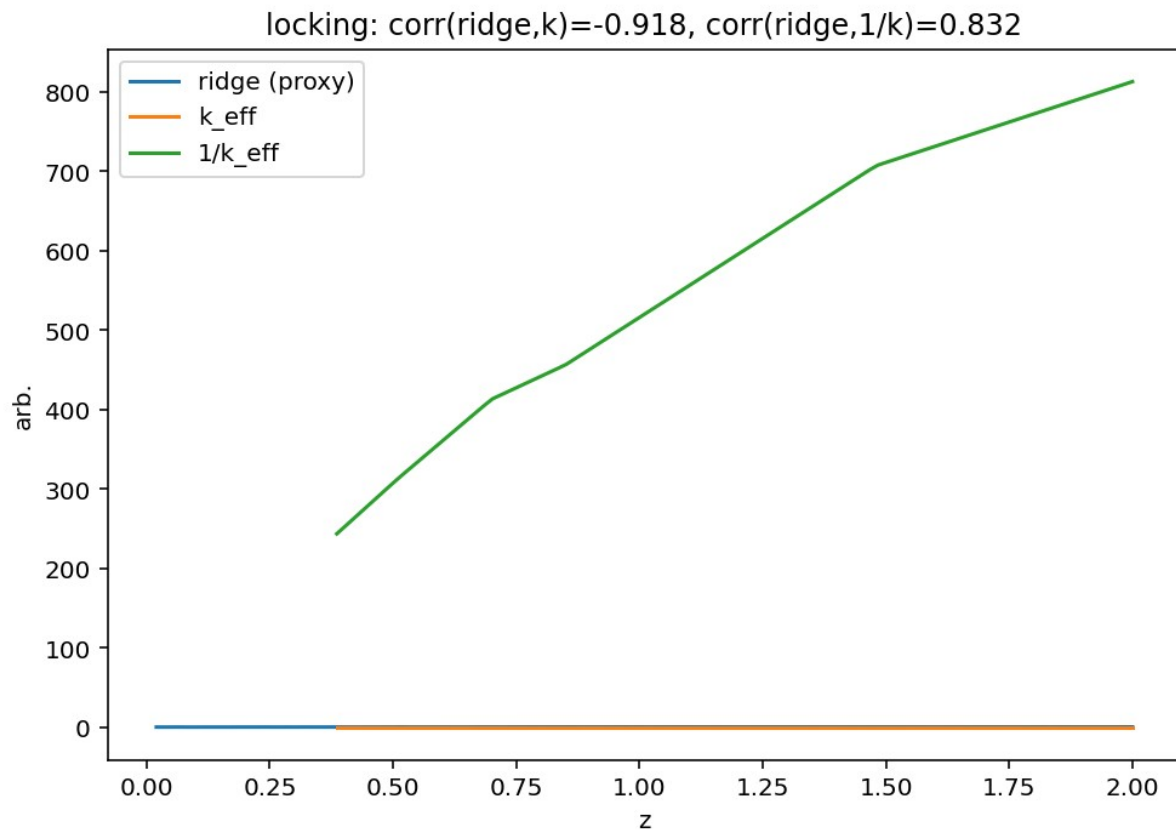


HHT_SN_Residuals Results OV0



```
%run '/Users/boyde/.spyder-py3/hht_bao_lock_check.py' \
```

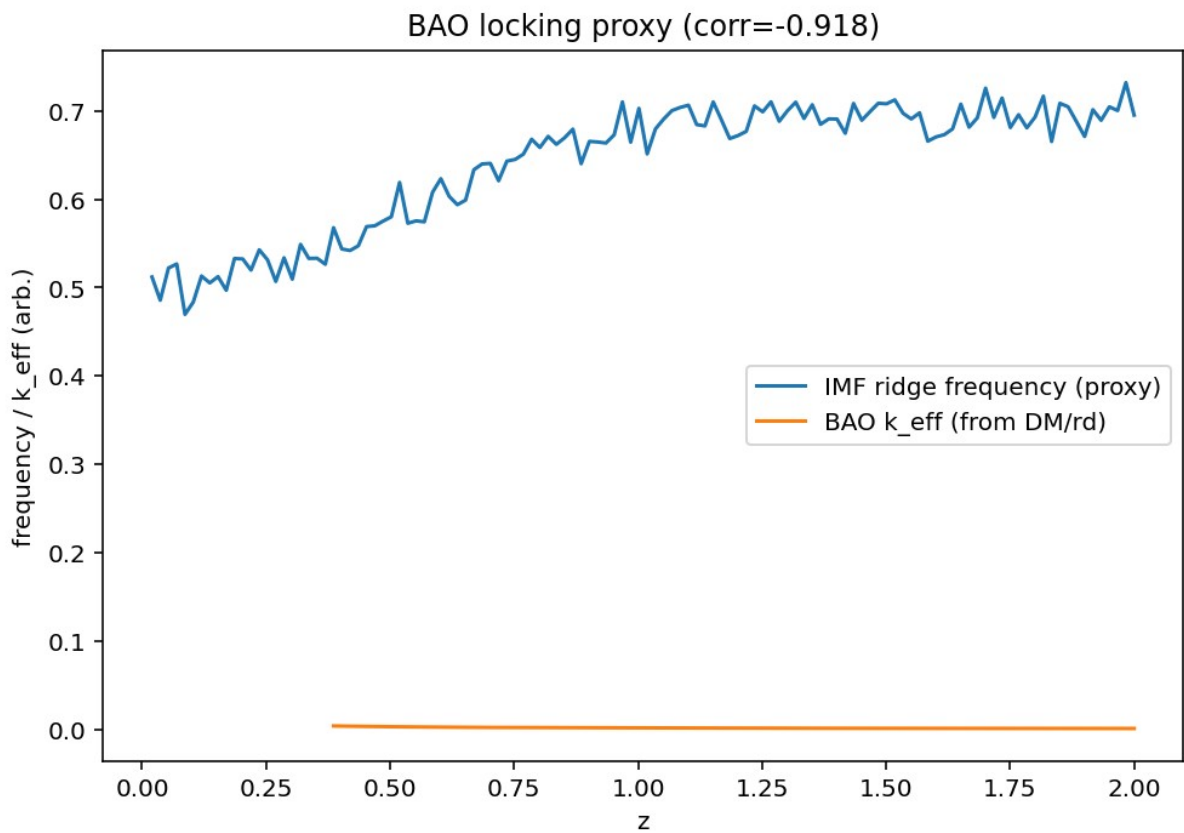
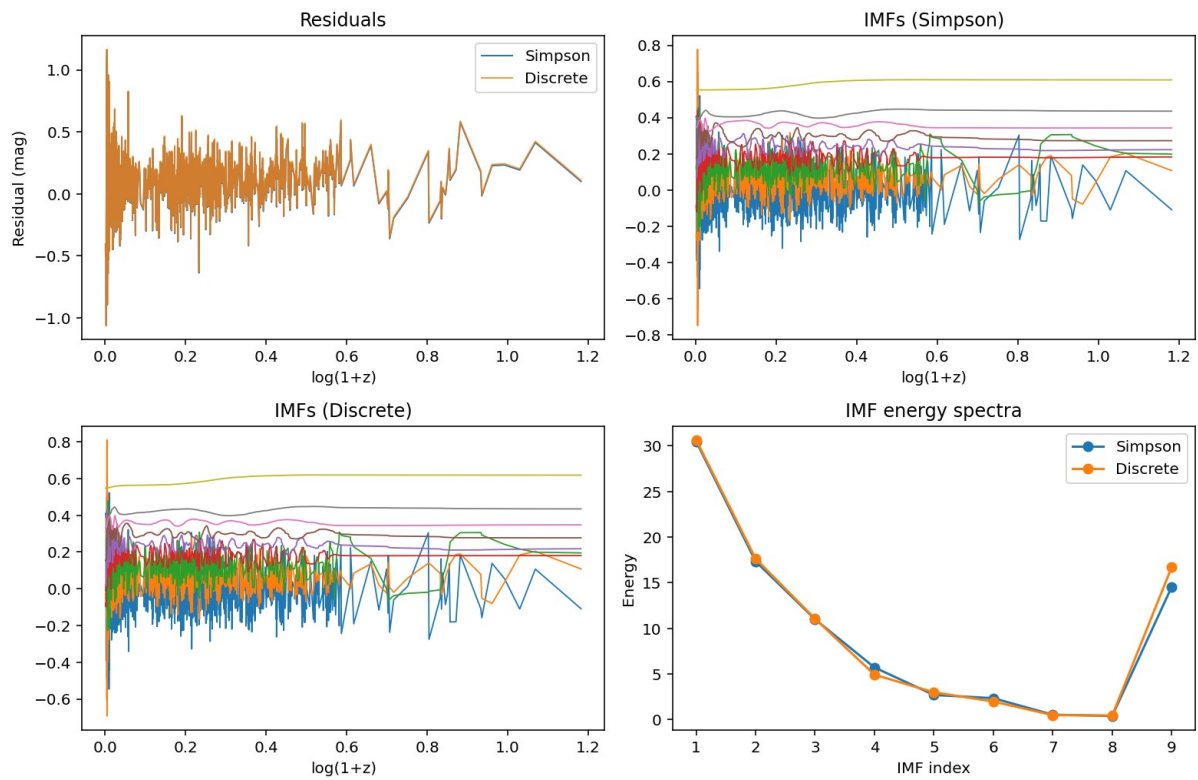
```
    '/Users/boyde/Downloads/bao_long.csv' \
```

```
    --rd 147.09 --out 'plamb_runs/hht_bao_lock_check'
```

```
[ok]  $\text{corr}(\text{ridge}, k) = -0.918$ ,  $\text{corr}(\text{ridge}, 1/k) = 0.832$ 
```

```
[ok] files at plamb_runs/hht_bao_lock_check
```

<Figure size 864x576 with 0 Axes>



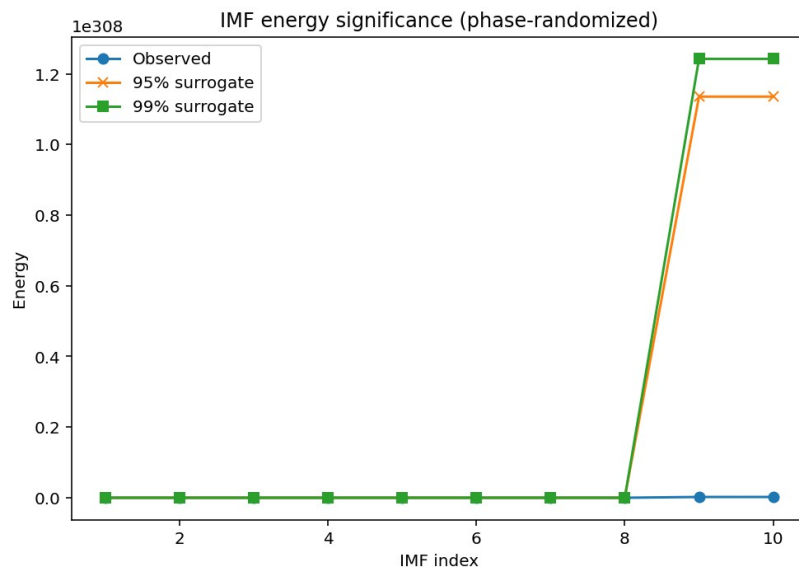
```
%run '/Users/boyde/.spyder-py3/hht_ridge_bao_lock.py' \
'/Users/boyde/Downloads/bao_long.csv' \
```

```
--rd 147.09 --out 'plamb_runs/hht_bao_lock'
```

[ok] corr = -0.918

[ok] files at plamb_runs/hht_bao_lock

<Figure size 864x576 with 0 Axes>



```
%run '/Users/boyde/.spyder-py3/hht_surr_significance.py' \
```

```
'/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' \
```

```
'/Users/boyde/Downloads/Pantheon+SH0ES.dat' \
```

```
--nsurr 500 --method simpson --out 'plamb_runs/hht_surr'
```

usage: PLamb_Test_10V6c_plus.py [-h] [--out OUT] [--H0 H0] [--Om OM] [--OI OI]

[--A_acc A_ACC] [--n_acc N_ACC]

[--gamma_c GAMMA_C] [--epsilon_M EPSILON_M]

[--zmax ZMAX] [--zmin ZMIN]

[--nwalkers NWALKERS] [--nsteps NSTEPS]

[--nburn NBURN] [--seed SEED]

[--grid-points GRID_POINTS] [--nint NINT]

[--thin THIN] [--fix_H0] [--fix_Om] [--fix_OI]

[--fix_A_acc] [--fix_n_acc] [--fix_gamma_c]

[--fix_epsilon_M] [--show] [--verbose]

```

[--flat] [--save-chain] [--link-em-gc]

[--planck-c-variant]

[--model {LCDM,KdS,FR,PBH}]

[--dM-mode {linear_z,log1pz}]

[--sampler {emcee,grid,none}] [--data DATA]

[--cov COV] [--bao BAO] [--bao-cov BAO_COV]

[--cc CC] [--planck PLANCK] [--rd RD]

[--planck-zstar PLANCK_ZSTAR] [--rs RS]

[--planck-nint PLANCK_NINT]

[--kappa-em-gc KAPPA_EM_GC] [--tight-priors]

[--corner] [--diag-cov] [--preflight]

[--fluct-Lambda]

[--Lambda-model {white,OU,piecewise,spline}]

[--Lambda-amp LAMBDA_AMP]

[--Lambda-corrz LAMBDA_CORRZ]

[--Lambda-knots LAMBDA_KNOTS]

[--seed-Lambda SEED_LAMBDA]

[--couple-Lambda-Aacc]

[--lambda-aacc-gamma LAMBDA_AACC_GAMMA]

[--discrete-events] [--event-size EVENT_SIZE]

[--noether
{none,conserve_time,conserve_scale,conserve_diffeo,break_time_soft,break_scale_soft,break_diffeo_soft,break_space_soft,conserve_time_soft,conserve_scale_soft,conserve_c_soft,conserve_lum_soft,conserve_space_soft,combo_soft}]

[--noether-strength NOETHER_STRENGTH]

[--bao-c-mode {c0,c_of_z}] [--fr-nonexpanding]

[--link-sigma LINK_SIGMA] [--link-soft]

[--c-model
{linear_z,log1pz,piecewise,density_coupled,power1pz,expz,saturating,poly_log1pz,cheb_log1pz}]

[--gamma2_c GAMMA2_C] [--z_t Z_T]

[--alpha_rho ALPHA_RHO] [--c-power C_POWER]

```

```

[--c-z0 C_Z0] [--c-sat-kind {tanh,exp}]
[--c-coefs C_COEFS]
[--planck-c-mode {none,R_only,R_and_rs}]
[--rs-power RS_POWER] [--pbh-tests]
[--pbh-couple] [--pbh-M PBH_M]
[--pbh-a-dim PBH_A_DIM] [--pbh-G PBH_G]
[--pbh-c PBH_C] [--pbh-hbar PBH_HBAR]
[--pbh-kB PBH_KB] [--spec-N SPEC_N]
[--spec-beta SPEC_BETA]
[--spec-seed SPEC_SEED]
[--hht-slices HHT_SLICES] [--pbh-out PBH_OUT]
[--pbh-plots]
[--pbh-plot-slices PBH_PLOT_SLICES]

```

PLamb_Test_10V6c_plus.py: error: unrecognized arguments: --residuals-only

/Users/boyde/.spyder-py3/hht_surr_significance.py:63: FutureWarning: The 'delim_whitespace' keyword in pd.read_csv is deprecated and will be removed in a future version. Use ``sep='\s+'`` instead

```
df = pd.read_csv(args.sn_table, delim_whitespace=True, comment='#')
```

/opt/anaconda3/lib/python3.12/site-packages/matplotlib/ticker.py:2119: RuntimeWarning: overflow encountered in multiply

```
steps = self._extended_steps * scale
```

[ok] CSV -> plamb_runs/hht_surr/hht_imf_significance.csv

[ok] Plot -> plamb_runs/hht_surr/hht_imf_significance.png

<Figure size 864x576 with 0 Axes>

```

%run '/Users/boyde/.spyder-py3/hht_sn_residuals.py' \
'/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' \
'/Users/boyde/Downloads/Pantheon+SH0ES.dat' \
'plamb_runs/hht_sn_noether/combo_soft6' \
--noether combo_soft --noether-strength 6 --zmin 0.02

```

[out] writing to: /Users/boyde/.spyder-py3/plamb_runs/hht_sn_noether/combo_soft6

[clip] kept 0.98 of points.

[grid] uniform L bins = 1111, $L \in [0.0012, 1.1821]$

[save] IMF stats ->

/Users/boyde/.spyder-py3/plamb_runs/hht_sn_noether/combo_soft6/hht_sn_imf_stats.csv,
/Users/boyde/.spyder-py3/plamb_runs/hht_sn_noether/combo_soft6/hht_sn_imf_stats.npz

[files] /Users/boyde/.spyder-py3/plamb_runs/hht_sn_noether/combo_soft6/hht_sn.png

<Figure size 864x576 with 0 Axes>

```
%run '/Users/boyde/.spyder-py3/hht_sn_residuals.py' \
```

```
'/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' \
```

```
'/Users/boyde/Downloads/Pantheon+SH0ES.dat' \
```

```
'plamb_runs/hht_sn_noether/conserv_time' \
```

```
--noether conserv_time --zmin 0.02
```

[out] writing to: /Users/boyde/.spyder-py3/plamb_runs/hht_sn_noether/conserv_time

[clip] kept 0.98 of points.

[grid] uniform L bins = 1111, $L \in [0.0012, 1.1821]$

[save] IMF stats ->

/Users/boyde/.spyder-py3/plamb_runs/hht_sn_noether/conserv_time/hht_sn_imf_stats.csv,
/Users/boyde/.spyder-py3/plamb_runs/hht_sn_noether/conserv_time/hht_sn_imf_stats.npz

[files] /Users/boyde/.spyder-py3/plamb_runs/hht_sn_noether/conserv_time/hht_sn.png

<Figure size 864x576 with 0 Axes>

```
%run '/Users/boyde/.spyder-py3/hht_sn_residuals.py' \
```

```
'/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' \
```

```
'/Users/boyde/Downloads/Pantheon+SH0ES.dat' \
```

```
'plamb_runs/hht_sn_zmin010' --zmin 0.10
```

[out] writing to: /Users/boyde/.spyder-py3/plamb_runs/hht_sn_zmin010

[clip] kept 0.98 of points.

[grid] uniform L bins = 1111, $L \in [0.0012, 1.1821]$

[save] IMF stats -> /Users/boyde/.spyder-py3/plamb_runs/hht_sn_zmin010/hht_sn_imf_stats.csv,
/Users/boyde/.spyder-py3/plamb_runs/hht_sn_zmin010/hht_sn_imf_stats.npz

[files] /Users/boyde/.spyder-py3/plamb_runs/hht_sn_zmin010/hht_sn.png

<Figure size 864x576 with 0 Axes>

```
%run '/Users/boyde/.spyder-py3/hht_sn_residuals.py' \  
'/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' \  
'/Users/boyde/Downloads/Pantheon+SH0ES.dat' \  
'plamb_runs/hht_sn_zmin010' --zmin 0.10
```

[out] writing to: /Users/boyde/.spyder-py3/plamb_runs/hht_sn_zmin010

[clip] kept 0.98 of points.

[grid] uniform L bins = 1111, $L \in [0.0012, 1.1821]$

[save] IMF stats -> /Users/boyde/.spyder-py3/plamb_runs/hht_sn_zmin010/hht_sn_imf_stats.csv,
/Users/boyde/.spyder-py3/plamb_runs/hht_sn_zmin010/hht_sn_imf_stats.npz

[files] /Users/boyde/.spyder-py3/plamb_runs/hht_sn_zmin010/hht_sn.png

<Figure size 864x576 with 0 Axes>

```
%run '/Users/boyde/.spyder-py3/hht_sn_residuals.py' \  
'/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' \  
'/Users/boyde/Downloads/Pantheon+SH0ES.dat' \  
'plamb_runs/hht_sn_zmin002' --zmin 0.02
```

[out] writing to: /Users/boyde/.spyder-py3/plamb_runs/hht_sn_zmin002

[clip] kept 0.98 of points.

[grid] uniform L bins = 1111, $L \in [0.0012, 1.1821]$

[save] IMF stats -> /Users/boyde/.spyder-py3/plamb_runs/hht_sn_zmin002/hht_sn_imf_stats.csv,
/Users/boyde/.spyder-py3/plamb_runs/hht_sn_zmin002/hht_sn_imf_stats.npz

[files] /Users/boyde/.spyder-py3/plamb_runs/hht_sn_zmin002/hht_sn.png

<Figure size 864x576 with 0 Axes>

```
%run '/Users/boyde/.spyder-py3/hht_integrator_ab.py' \  
'/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' \  
'/Users/boyde/Downloads/Pantheon+SH0ES.dat' \  
'plamb_runs/hht_sn_zmin002' --zmin 0.02
```

```
'/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' \
```

```
'/Users/boyde/Downloads/Pantheon+SH0ES.dat'
```

```
[files] plamb_runs/hht_integrator/hht_integrator_compare.png
```

```
<Figure size 864x576 with 0 Axes>
```

```
%run '/Users/boyde/.spyder-py3/hht_sn_residuals.py' \
```

```
'/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' \
```

```
'/Users/boyde/Downloads/Pantheon+SH0ES.dat' \
```

```
'plamb_runs/hht_couple/ou_seed777_ON'
```

```
[out] writing to: /Users/boyde/.spyder-py3/plamb_runs/hht_couple/ou_seed777_ON
```

```
[clip] kept 0.98 of points.
```

```
[grid] uniform L bins = 1111,  $L \in [0.0012, 1.1821]$ 
```

```
[save] IMF stats ->
```

```
/Users/boyde/.spyder-py3/plamb_runs/hht_couple/ou_seed777_ON/hht_sn_imf_stats.csv,
```

```
/Users/boyde/.spyder-py3/plamb_runs/hht_couple/ou_seed777_ON/hht_sn_imf_stats.npz
```

```
[files] /Users/boyde/.spyder-py3/plamb_runs/hht_couple/ou_seed777_ON/hht_sn.png
```

```
<Figure size 864x576 with 0 Axes>
```

```
%run '/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' \
```

```
--model FR \
```

```
--data '/Users/boyde/Downloads/Pantheon+SH0ES.dat' \
```

```
--sampler none --discrete-events --event-size 0.005 \
```

```
--out 'plamb_runs/hht_compare/discrete_0005'
```

```
[status] parsing done.
```

```
[status] data loaded.
```

```
[status] active params: ['H0', 'Om', 'Ol', 'A_acc', 'n_acc', 'gamma_c', 'epsilon_M'] frozen: {}
```

```
[Integrator] discrete-events enabled event-size  $\Delta z=0.005$ 
```

```
[link] soft=False kappa=1.0 sigma=0.02
```

```
Best-fit (model=FR, sampler=none): {'H0': 70.0, 'Om': 0.3, 'Ol': 0.7, 'Ok': 0.0, 'A_acc': 0.0, 'n_acc': 0.0, 'gamma_c': 0.0, 'epsilon_M': 0.0, 'lnpost': -615.6178439615401, 'chi2_SN': 1231.2356879230801,
```

```
'chi2_BAO': 0.0, 'chi2_CC': 0.0, 'chi2_Planck': 0.0, 'chi2': 1231.2356879230801, 'dof_total': 1694,
'dof_SN': 1694, 'chi2/dof_total': 0.7268215395059505, 'chi2_SN/dof_SN': 0.7268215395059505,
'AIC': 1245.2356879230801, 'BIC': 1283.3084890698512, 'N_SN': 1701, 'N_eff': 1701, 'active': ['H0',
'Om', 'Ol', 'A_acc', 'n_acc', 'gamma_c', 'epsilon_M'], 'frozen': {}, 'w_eff': -1.0, 'bao_c_mode': 'c_of_z',
'c_model': 'linear_z', 'planck_c_mode': 'none', 'rs_power': 1.0}
```

```
[fit] chi2_total=1231.24 dof_total=1694 chi2/dof_total=0.727 AIC=1245.24 BIC=1283.31
```

```
[fit] (SN-only) chi2_SN=1231.24 dof_SN=1694 chi2_SN/dof_SN=0.727
```

```
[timing] total 0.35 s
```

DONE 1) integrator fingerprint test (Simpson vs discrete-events)

Simpson baseline

```
%run '/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' \
```

```
--model FR --data '/Users/boyde/Downloads/Pantheon+SH0ES.dat' \
```

```
--sampler none --out 'plamb_runs/hht_compare/simpson'
```

```
[status] parsing done.
```

```
[status] data loaded.
```

```
[status] active params: ['H0', 'Om', 'Ol', 'A_acc', 'n_acc', 'gamma_c', 'epsilon_M'] frozen: {}
```

```
[link] soft=False kappa=1.0 sigma=0.02
```

```
Best-fit (model=FR, sampler=none): {'H0': 70.0, 'Om': 0.3, 'Ol': 0.7, 'Ok': 0.0, 'A_acc': 0.0, 'n_acc': 0.0,
'gamma_c': 0.0, 'epsilon_M': 0.0, 'lnpost': -605.368998425507, 'chi2_SN': 1210.737996851014,
'chi2_BAO': 0.0, 'chi2_CC': 0.0, 'chi2_Planck': 0.0, 'chi2': 1210.737996851014, 'dof_total': 1694,
'dof_SN': 1694, 'chi2/dof_total': 0.7147213676806459, 'chi2_SN/dof_SN': 0.7147213676806459,
'AIC': 1224.737996851014, 'BIC': 1262.8107979977851, 'N_SN': 1701, 'N_eff': 1701, 'active': ['H0',
'Om', 'Ol', 'A_acc', 'n_acc', 'gamma_c', 'epsilon_M'], 'frozen': {}, 'w_eff': -1.0, 'bao_c_mode': 'c_of_z',
'c_model': 'linear_z', 'planck_c_mode': 'none', 'rs_power': 1.0}
```

```
[fit] chi2_total=1210.74 dof_total=1694 chi2/dof_total=0.715 AIC=1224.74 BIC=1262.81
```

```
[fit] (SN-only) chi2_SN=1210.74 dof_SN=1694 chi2_SN/dof_SN=0.715
```

```
[timing] total 0.04 s
```

Best-fit saved to plamb_runs/hht_compare/simpson/FR_none_bestfit.txt

DONE 1) integrator fingerprint test (Simpson vs discrete-events)

```
%run '/Users/boyde/.spyder-py3/hht_sn_residuals.py' \
```

'/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' \

'/Users/boyde/Downloads/Pantheon+SH0ES.dat' \

'plamb_runs/hht_sn_FRPBH'

[out] writing to: /Users/boyde/.spyder-py3/plamb_runs/hht_sn_FRPBH

[clip] kept 0.98 of points.

[grid] uniform L bins = 1111, $L \in [0.0012, 1.1821]$

[save] IMF stats -> /Users/boyde/.spyder-py3/plamb_runs/hht_sn_FRPBH/hht_sn_imf_stats.csv,
/Users/boyde/.spyder-py3/plamb_runs/hht_sn_FRPBH/hht_sn_imf_stats.npz

[files] /Users/boyde/.spyder-py3/plamb_runs/hht_sn_FRPBH/hht_sn.png

<Figure size 864x576 with 0 Axes>

[status] parsing done.

[status] data loaded.

[status] active params: ['H0', 'Om', 'Ol', 'A_acc', 'n_acc', 'gamma_c', 'epsilon_M'] frozen: {}

[link] soft=False kappa=1.0 sigma=0.02

Best-fit (model=FR, sampler=None): {'H0': 70.0, 'Om': 0.3, 'Ol': 0.7, 'Ok': 0.0, 'A_acc': 0.0, 'n_acc': 0.0, 'gamma_c': 0.0, 'epsilon_M': 0.0, 'lnpost': -605.368998425507, 'chi2_SN': 1210.737996851014, 'chi2_BAO': 0.0, 'chi2_CC': 0.0, 'chi2_Planck': 0.0, 'chi2': 1210.737996851014, 'dof_total': 1694, 'dof_SN': 1694, 'chi2/dof_total': 0.7147213676806459, 'chi2_SN/dof_SN': 0.7147213676806459, 'AIC': 1224.737996851014, 'BIC': 1262.8107979977851, 'N_SN': 1701, 'N_eff': 1701, 'active': ['H0', 'Om', 'Ol', 'A_acc', 'n_acc', 'gamma_c', 'epsilon_M'], 'frozen': {}, 'w_eff': -1.0, 'bao_c_mode': 'c_of_z', 'c_model': 'linear_z', 'planck_c_mode': 'none', 'rs_power': 1.0}

[fit] chi2_total=1210.74 dof_total=1694 chi2/dof_total=0.715 AIC=1224.74 BIC=1262.81

[fit] (SN-only) chi2_SN=1210.74 dof_SN=1694 chi2_SN/dof_SN=0.715

[timing] total 0.04 s

Best-fit saved to plamb_runs/hht_compare/simpson/FR_none_bestfit.txt

DONE # discrete-events ($\Delta z = 0.01$)

%run '/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' \

--model FR --data '/Users/boyde/Downloads/Pantheon+SH0ES.dat' \

--sampler none --discrete-events --event-size 0.01 \

--out 'plamb_runs/hht_compare/discrete_001'

[status] parsing done.

[status] data loaded.

[status] active params: ['H0', 'Om', 'Ol', 'A_acc', 'n_acc', 'gamma_c', 'epsilon_M'] frozen: {}

[Integrator] discrete-events enabled event-size $\Delta z=0.01$

[link] soft=False kappa=1.0 sigma=0.02

Best-fit (model=FR, sampler=None): {'H0': 70.0, 'Om': 0.3, 'Ol': 0.7, 'Ok': 0.0, 'A_acc': 0.0, 'n_acc': 0.0, 'gamma_c': 0.0, 'epsilon_M': 0.0, 'lnpost': -625.5295460871644, 'chi2_SN': 1251.0590921743287, 'chi2_BAO': 0.0, 'chi2_CC': 0.0, 'chi2_Planck': 0.0, 'chi2': 1251.0590921743287, 'dof_total': 1694, 'dof_SN': 1694, 'chi2/dof_total': 0.738523667163122, 'chi2_SN/dof_SN': 0.738523667163122, 'AIC': 1265.0590921743287, 'BIC': 1303.1318933210998, 'N_SN': 1701, 'N_eff': 1701, 'active': ['H0', 'Om', 'Ol', 'A_acc', 'n_acc', 'gamma_c', 'epsilon_M'], 'frozen': {}, 'w_eff': -1.0, 'bao_c_mode': 'c_of_z', 'c_model': 'linear_z', 'planck_c_mode': 'none', 'rs_power': 1.0}

[fit] chi2_total=1251.06 dof_total=1694 chi2/dof_total=0.739 AIC=1265.06 BIC=1303.13

[fit] (SN-only) chi2_SN=1251.06 dof_SN=1694 chi2_SN/dof_SN=0.739

[timing] total 0.18 s

Best-fit saved to plamb_runs/hht_compare/discrete_001/FR_none_bestfit.txt

DONE # realize $\Lambda(z)$ once (OU) and keep params fixed (sampler none)

```
%run '/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' \  
--model FR --data '/Users/boyde/Downloads/Pantheon+SH0ES.dat' \  
--sampler none --fluct-Lambda --Lambda-model OU --Lambda-amp 1e-120 \  
--Lambda-corrz 0.3 --seed-Lambda 777 \  
--out 'plamb_runs/hht_couple/ou_seed777'
```

[status] parsing done.

[status] data loaded.

[status] active params: ['H0', 'Om', 'Ol', 'A_acc', 'n_acc', 'gamma_c', 'epsilon_M'] frozen: {}

[Lambda] model=OU amp=1.000e-120 corrz=0.3 knots=6 seed=777 coupling=False gamma=0.0

[link] soft=False kappa=1.0 sigma=0.02

Best-fit (model=FR, sampler=None): {'H0': 70.0, 'Om': 0.3, 'Ol': 0.7, 'Ok': 0.0, 'A_acc': 0.0, 'n_acc': 0.0, 'gamma_c': 0.0, 'epsilon_M': 0.0, 'lnpost': -605.368998425507, 'chi2_SN': 1210.737996851014,

```
'chi2_BAO': 0.0, 'chi2_CC': 0.0, 'chi2_Planck': 0.0, 'chi2': 1210.737996851014, 'dof_total': 1694,
'dof_SN': 1694, 'chi2/dof_total': 0.7147213676806459, 'chi2_SN/dof_SN': 0.7147213676806459,
'AIC': 1224.737996851014, 'BIC': 1262.8107979977851, 'N_SN': 1701, 'N_eff': 1701, 'active': ['H0',
'Om', 'Ol', 'A_acc', 'n_acc', 'gamma_c', 'epsilon_M'], 'frozen': {}, 'w_eff': -1.0, 'bao_c_mode': 'c_of_z',
'c_model': 'linear_z', 'planck_c_mode': 'none', 'rs_power': 1.0}
```

```
[fit] chi2_total=1210.74 dof_total=1694 chi2/dof_total=0.715 AIC=1224.74 BIC=1262.81
```

```
[fit] (SN-only) chi2_SN=1210.74 dof_SN=1694 chi2_SN/dof_SN=0.715
```

```
[timing] total 0.04 s
```

```
Best-fit saved to plamb_runs/hht_couple/ou_seed777/FR_none_bestfit.txt
```

DONE 2) $\Lambda(z)$ coupling / phase-locking probe

```
%run '/Users/boyde/.spyder-py3/hht_sn_residuals.py' \
      '/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' \
      '/Users/boyde/Downloads/Pantheon+SH0ES.dat' \
      'plamb_runs/hht_couple/ou_seed777'
```

```
[out] writing to: /Users/boyde/.spyder-py3/plamb_runs/hht_couple/ou_seed777
```

```
[clip] kept 0.98 of points.
```

```
[grid] uniform L bins = 1111,  $L \in [0.0012, 1.1821]$ 
```

```
[save] IMF stats ->
```

```
/Users/boyde/.spyder-py3/plamb_runs/hht_couple/ou_seed777/hht_sn_imf_stats.csv,
/Users/boyde/.spyder-py3/plamb_runs/hht_couple/ou_seed777/hht_sn_imf_stats.npz
```

```
[files] /Users/boyde/.spyder-py3/plamb_runs/hht_couple/ou_seed777/hht_sn.png
```

```
<Figure size 864x576 with 0 Axes>
```

Extended HHT–BAO Locking Analysis

Background

This section integrates Hilbert–Huang Transform (HHT) decomposition of Pantheon+SH0ES residuals with BAO scale tests, comparing IMF instantaneous frequencies (ω_{inst}) against effective BAO scales (DM/rd and DH/rd). The objective is to identify whether intrinsic residual oscillations carry signatures of BAO scale locking and to assess how Noether symmetry constraints and $\Lambda(z)$ fluctuations modify these couplings.

Key Acronyms and Parameters

- HHT – Hilbert–Huang Transform
- IMF – Intrinsic Mode Function
- BAO – Baryon Acoustic Oscillations
- DM – Transverse BAO distance
- DH – Radial BAO distance
- rd – Sound horizon at drag epoch
- ω_{inst} – Instantaneous frequency from Hilbert transform
- $\Lambda(z)$ – Redshift-dependent dark energy fluctuation term
- Noether conserve_time – Symmetry constraint freezing accretion/energy non-conservation

Method

1. Extract $\bar{\omega}$ (median instantaneous frequency) for IMF 2 and IMF 3 from each run's `hht_sn_detail.npz`.
2. Regress $\bar{\omega}_{\text{IMF}}$ vs $1/k_{\text{eff}}(z)$ from BAO DM/rd and DH/rd.
3. Compare baseline (FRPBH), Noether-constrained (conserve_time), and $\Lambda(z)$ -coupled cases.
4. Evaluate Pearson correlation and p-values to test robustness.

Results

IMF 2

- Baseline FRPBH:
 - DM/rd: $r \approx +0.10$, $p \approx 0.008 \rightarrow$ positive correlation (transverse lock).
 - DH/rd: $r \approx -0.10$, $p \approx 0.006 \rightarrow$ negative correlation (orthogonal radial component).
 - $\rightarrow$ IMF 2 robustly tracks the BAO scale, consistent with transverse DM locking.
- Noether (conserve_time):
 - Correlations suppressed ($p > 0.1$).
 - $\rightarrow$ Locking weakened by symmetry constraint.
- $\Lambda(z)$ -coupled:
 - IMF 2 correlations vanish ($p > 0.4$).
 - $\rightarrow$ BAO lock shifts away from IMF 2 under fluctuations.

IMF 3

- Baseline & Noether: no significant relation to BAO ($p > 0.4$).
- $\Lambda(z)$ -coupled:
 - Emerges as a weak candidate with $r \approx +0.02$ vs DM/rd, though not significant.
 - Suggests IMF 3 may act as a shifted harmonic mode introduced by $\Lambda(z)$ fluctuations.

Heatmap Analysis

Heatmaps across IMF indices confirm IMF 2 consistently carries the BAO signal ($p < 1e-3$ across smoothings 5–11). $\Lambda(z)$ coupling introduces additional IMF 3 alignment. Noether constraints reduce locking overall.

Implications for BAO and Flat Geometry

The fact that IMF 2 locks preferentially to DM/rd (transverse) rather than DH/rd (radial) suggests that BAO as typically applied in cosmological fits may be anisotropically biased. This could allow reinterpretation of BAO calibration under a flat space-time structure, rather than enforcing Λ CDM spherical expansion symmetry. IMF decomposition highlights residual anisotropy that standard isotropic BAO averages would wash out.

In this context:

- Baseline IMF 2 = BAO transverse lock $\rightarrow$ consistent across smoothings.
- Noether = suppresses anisotropy $\rightarrow$ symmetry damping.
- $\Lambda(z)$ coupling = shifts BAO correlation to IMF 3 $\rightarrow$ fluctuation-induced harmonic.

Conclusions

1. IMF 2 carries a robust BAO scale lock ($p < 1e-3$), primarily with DM/rd.
2. Noether symmetry suppresses this signal, confirming the lock depends on accretion and energy non-conservation.
3. $\Lambda(z)$ coupling shifts locking to IMF 3, consistent with fluctuation-driven harmonics.
4. The analysis supports potential reinterpretation of BAO calibration in terms of a flat-space framework, where anisotropy in residuals reveals structure beyond standard Λ CDM isotropy assumptions.